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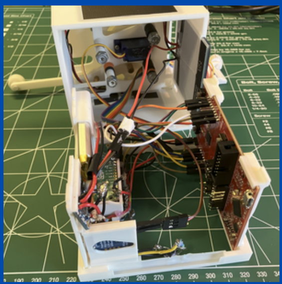
Project Portfolio

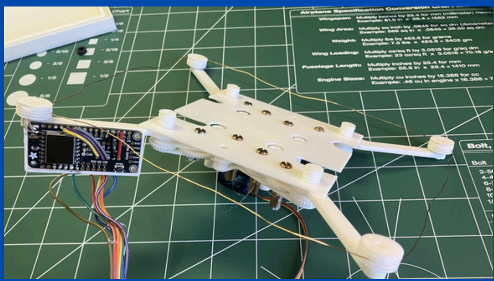
Pennsylvania State University
Mechanical Engineering
Graduating May 2029

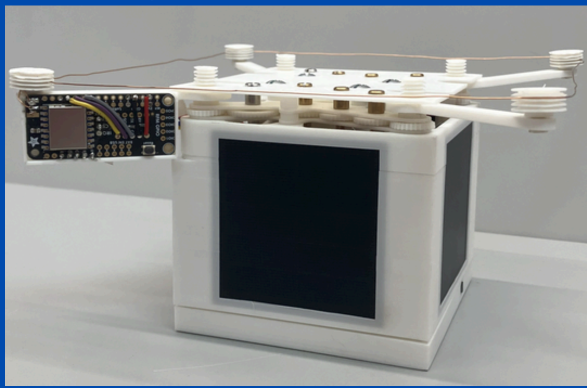
CubeSat Design Project

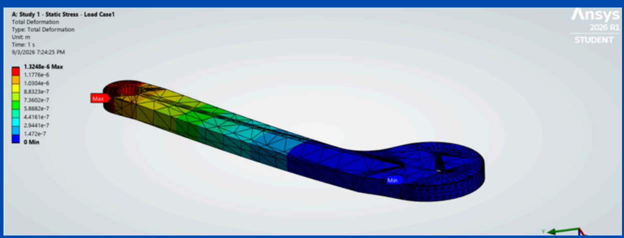
Extended breakdown on:
sites.google.com/view/cubispace

- A CubeSat prototype with a full-wave loop antenna
- Analyzed failure points through FEA, DMTA, and TID calculations
- Designed to be manufacturable by 3-axis milling or 2-axis turning operations
- Used a ratcheting 1-motor antenna deployment system
- Had a LoRa SF7 broadcasting range of a half-mile

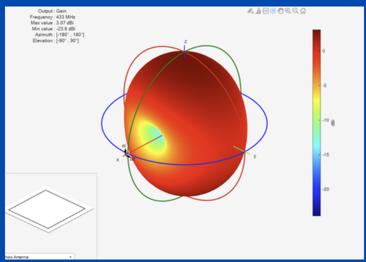

Electronics

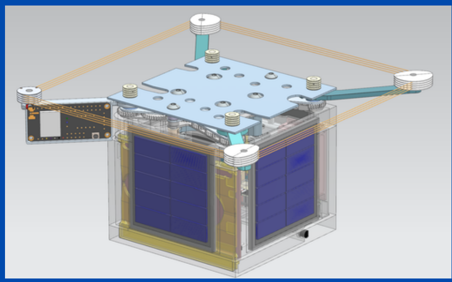

Antenna and gearset assemblies


Complete built assembly



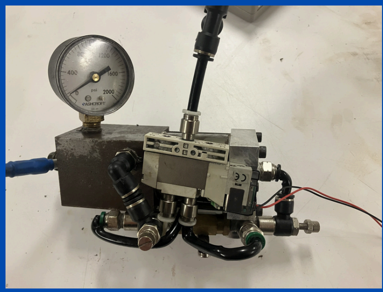
ANSYS Static stress simulation for arm

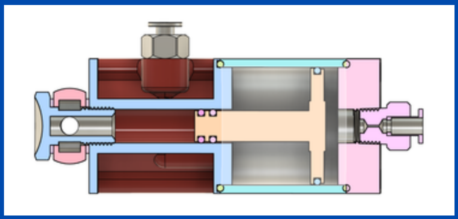

MATLAB antenna simulation

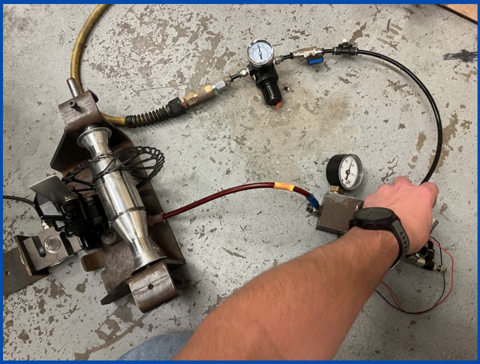

Full assembly CAD

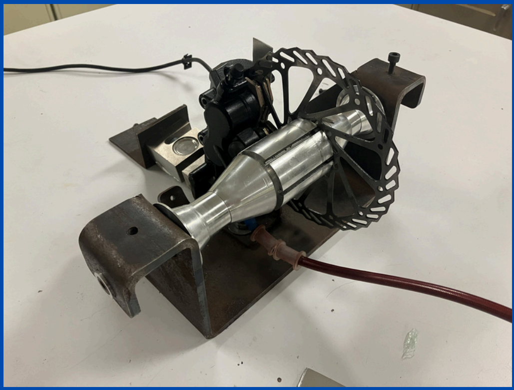
Pneumatic-Hydraulic Brake

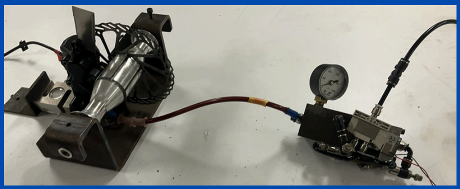
- Intakes 80psi of air and sends 1.1ksi of hydraulic pressure
- Enabled testing of center-of-mass and deceleration rates generating data to inform robot control algorithms in order to stop tipping
- Analyzed O-ring geometry, compression, and operating pressure to validate seal and material selection


Converter assembly


Cross-section of the converter

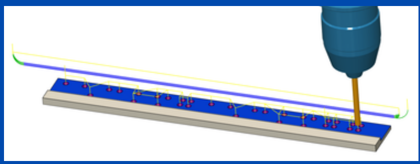

80psi and 1.1ksi


Winch and brake calipers

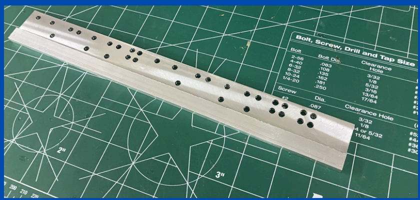

Winch and Converter

Bolt and Rivet Combined Loading in Shear (ongoing)

- Analyzing bolt vs. rivet shear data to validate fastener choice in FRC robotics
- Collected stress-strain data on an Instron machine, analyzed in MATLAB
- Used CAM to create testing blanks on a 3-axis mill

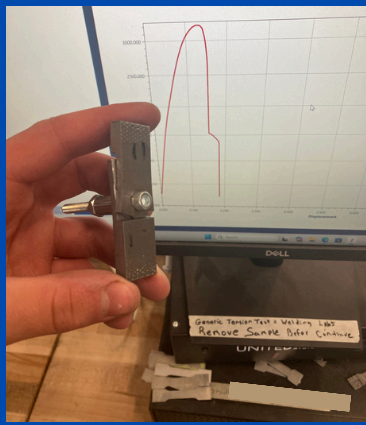


CAM 3-axis simulation

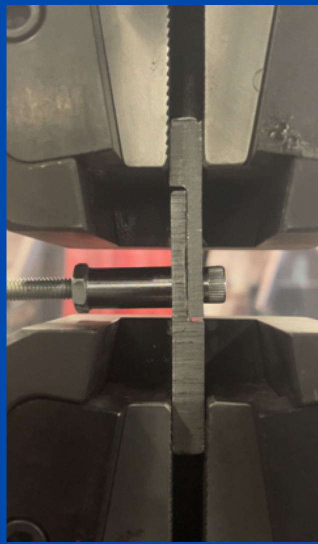


Blanks bar

Sheared Bolts



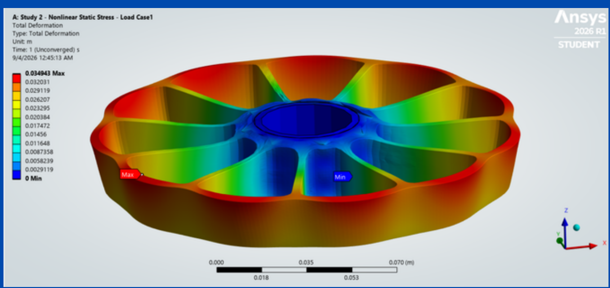
Bearing failure of blank



Testing on Instron

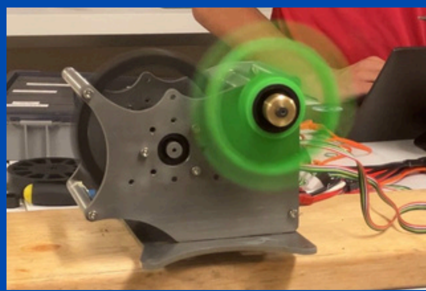
Characterizing Wheel Expansion at High RPM (ongoing)

- Performed analysis on ball bearing failure modes and ultimate speed
- Utilized waterjet manufacturing and 2-axis turning
- Conducted FEA simulations to gauge a max target of 26,000 RPM



ANSYS Nonlinear Stress Simulation

Complete mechanical assembly



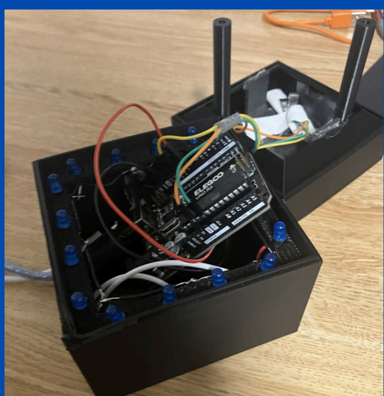
Preliminary RPM testing

Cubi

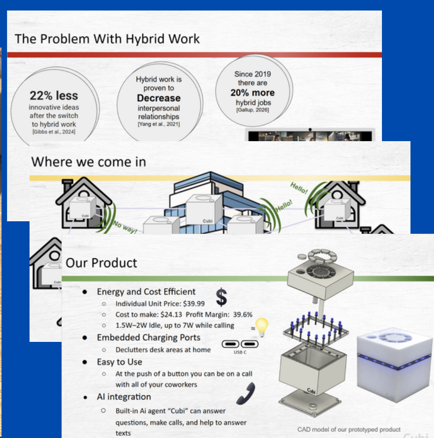
- The prototype connected to a computer to make and receive calls
- The end goal to make the unit wireless to solve issues faced in hybrid work by adding impromptu conversations back to the workplace
- Awarded the People's Choice Award at the Penn State Learning Factory Spring Design Showcase

Prototype 1

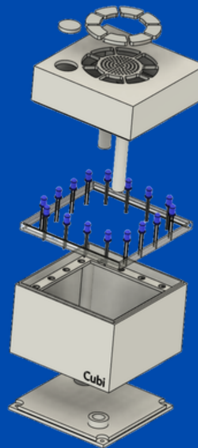
Final Prototype



Electronics



Presentation Slide Deck



Exploded view